NORMALITY ASCENDS ALONG A REGULAR MAP FROM A NORMAL NOETHERIAN RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- A Noetherian ring is normal if and only if it satisfies (S_2) and (R_1). Algebra, Lemma \ref{algebra-lemma-criterion-normal}, cited at proof lines 10564-10566. Replaces normality by the two local conditions.
- The conditions (S_k) and (R_k) ascend along regular ring maps. Algebra, Lemma \ref{algebra-lemma-Sk-goes-up} and Algebra, Lemma \ref{algebra-lemma-Rk-goes-up}, cited at proof lines 10567-10569. Give both conditions for the target.

Source reference. Original lemma 0BFK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14388–14388. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 031S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3995–3995. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0339: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4048–4048. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 033A: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4049–4049. Navigation pointer to a statement the

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $\varphi : R \to S$ be a ring map. Assume*

(1) *φ is regular,*
(2) *S is Noetherian, and*
(3) *R is Noetherian and normal.*

Then S is normal.

Proof. For Noetherian rings being normal is the same as having properties (S_2) and (R_1), see Algebra, Lemma 031S. Hence we may apply Algebra, Lemmas 0339 and 033A. □